# THE POLYGONAL BIGRAPHS $P_{8s,4s-2}$ ARE 2-EDGE-HAMILTONIAN

Abstract. Simmons conjectured that every polygonal bigraph $P_{m,k}$ is 2-edge-Hamiltonian. We prove the conjecture for the infinite family $P_{8s,4s-2}$, $s \geq 1$, by an explicit four-edge switch and cyclic rotations; the cases not already covered by Simmons are exactly $s \equiv 1 \pmod 3$ with $s \geq 4$. For those parameters the graphs lie in the arithmetic residue $2 < \gcd(m+1, k+1) < k+1$ isolated in [1, p. 109]. The first such case is $P_{32,14}$; the table of the addendum [2] leaves that cell blank, so our theorem supplies the one case needed for that addendum's assertion that all $P_{m,k}$ on 66 or fewer vertices are 2-edge-Hamiltonian. The remaining parameters are not addressed.

## 1. The construction

A graph is *2-edge-Hamiltonian* if every pair of distinct edges lies on a common Hamilton cycle. Simmons defined $P_{m,k}$ from a cycle on $2m$ vertices by joining each odd vertex, in a one-based clockwise labeling, to the even vertex $2k+1$ positions ahead [1, pp. 101–102]. He conjectured that every $P_{m,k}$ is 2-edge-Hamiltonian [1, p. 101]. The map $v \mapsto -v \pmod{2m}$ is an isomorphism between Simmons's orientation and the one used below, in which each even vertex is joined to the odd vertex $2k+1$ positions ahead, so we may work with the latter.

**Theorem 1.** *For every integer $s \geq 1$, the polygonal bigraph $P_{8s,4s-2}$ is 2-edge-Hamiltonian.*

*Proof.* Put

$$m = 8s, \qquad k = 4s - 2 = \frac{m}{2} - 2.$$

Work modulo $2m$ on vertices and modulo $m$ on subscripts. The three natural perfect matchings of $P_{m,k}$ are

$$A_i = \{2i, 2i+1\}, \qquad B_i = \{2i+1, 2i+2\}, \qquad C_i = \{2i, 2i+2k+1\}.$$

Write $A = \{A_i : i \in \mathbb{Z}/m\mathbb{Z}\}$, and similarly for $B$ and $C$.

The outer cycle $A \cup B$ is Hamilton. The subgraph $B \cup C$ is also Hamilton, because a chord followed by a $B$-edge sends the even subscript

$$i \longmapsto i + k + 1,$$

and

$$\gcd(k+1, m) = \gcd(4s-1, 8s) = 1.$$

2020 *Mathematics Subject Classification.* 05C45.

*Key words and phrases.* Hamilton cycle, polygonal bigraph, 2-edge-Hamiltonian.

These two Hamilton cycles cover every edge pair except pairs of the form $\{C_p, A_q\}$.

A chord followed by an $A$-edge sends the even subscript $i$ to $i+k$. Since

$$\gcd(k,m) = \gcd(4s-2, 8s) = 2,$$

the subgraph $A \cup C$ consists of two cycles, one on the even subscripts and one on the odd subscripts. Define

$$\begin{aligned} H = {} & (A \setminus \{A_k, A_{k+m/2}\}) \cup (C \setminus \{C_{k+1}, C_{k+1+m/2}\}) \\ & \cup \{B_{k-1}, B_k, B_{k-1+m/2}, B_{k+m/2}\}. \end{aligned}$$

We show that $H$ is a Hamilton cycle. For $\varepsilon \in \{0,1\}$, put $i_t = \varepsilon + tk$ in $\mathbb{Z}/m\mathbb{Z}$. The corresponding component of $A \cup C$ has cyclic order

$$(2i_0, 2i_1+1, 2i_1, 2i_2+1, \ldots, 2i_{m/2-1}, 2i_0+1).$$

Since

$$\frac{m}{4}k \equiv \frac{m}{2} \pmod{m},$$

the two deleted edges in each component occur halfway around that component. Reading the two cyclic orders gives four spanning paths with endpoint pairs

$$\begin{aligned} &\{2k+1, 2k+m\}, \quad \{2k, 2k+m+1\}, \\ &\{2k+2, 2k-1\}, \quad \{2k+m-1, 2k+m+2\}. \end{aligned}$$

Here the second line uses $k = m/2 - 2$, so

$$4k+3 = 2k+m-1, \qquad 4k+m+3 \equiv 2k-1 \pmod{2m}.$$

The four added edges join the paths in the cyclic order

$$\begin{aligned} 2k+1 &\xrightarrow{B_k} 2k+2 \rightsquigarrow 2k-1 \xrightarrow{B_{k-1}} 2k \rightsquigarrow 2k+m+1 \\ &\xrightarrow{B_{k+m/2}} 2k+m+2 \rightsquigarrow 2k+m-1 \xrightarrow{B_{k-1+m/2}} 2k+m \rightsquigarrow 2k+1. \end{aligned}$$

Thus $H$ is spanning, connected, and 2-regular, hence Hamiltonian.

Now let $C_p$ and $A_q$ be arbitrary and put $d = q - p \in \mathbb{Z}/m\mathbb{Z}$. Choose

$$i \notin \{k+1, k+1+m/2, k-d, k+m/2-d\}.$$

Such an $i$ exists because $m \geq 8$. Then $H$ contains both $C_i$ and $A_{i+d}$. The rotation $\rho(v) = v+2$ is an automorphism satisfying

$$\rho^r(C_i) = C_{i+r}, \qquad \rho^r(A_j) = A_{j+r}.$$

Taking $r = p - i$, the Hamilton cycle $\rho^r(H)$ contains $C_p$ and $A_{p+d} = A_q$. Hence every remaining edge pair lies on a Hamilton cycle. □

**Corollary 2.** *For every integer $t \geq 1$, the graph $P_{24t+8,12t+2}$ is 2-edge-Hamiltonian and satisfies*

$$2 < \gcd(m+1, k+1) < k+1.$$

*In particular, the first member is $P_{32,14}$.*

*Proof.* Set $s = 3t + 1$ in Theorem 1. Then

$$m = 24t + 8, \qquad k = 12t + 2, \qquad \gcd(m + 1, k + 1) = 3.$$

Since $k+1 = 12t+3 > 3$, the strict inequalities follow. This is precisely the residue $2 < \gcd(m+1, k+1) < k+1$ left open in [1, p. 109] and re-examined in [2, p. 102]. □

**Proposition 3.** *For every $s \geq 1$, the graph $P_{8s,4s-2}$ is not isomorphic to $P_{8s,k'}$ for any $k' \neq 4s - 2$ in the canonical range $1 \leq k' \leq 4s - 1$.*

*Proof.* For fixed $m$, the bipartite adjacency block of $P_{m,k}$ is a weight-three circulant with support affinely equivalent to

$$S_k = \{0, 1, k + 1\} \subseteq \mathbb{Z}/m\mathbb{Z}.$$

An isomorphism between two connected bipartite graphs either preserves or interchanges their parts; interchanging the parts transposes the block and replaces its support by its negative. It follows from [4, Theorem 1.1] that isomorphic polygonal bigraphs have affinely equivalent supports.

For the family above, put

$$a = k + 1 = \frac{m}{2} - 1.$$

Then $a$ is a unit modulo $m$, $a - 1$ is not, and

$$a^2 \equiv 1 \pmod{m}.$$

Consequently, after normalizing an ordered pair at unit distance to $0, 1$, the third element is one of

$$a, \qquad 1 - a, \qquad a^{-1}, \qquad 1 - a^{-1}.$$

Since $a^{-1} = a$, only $a$ and $1 - a$ occur. If $S_{k'} = \{0, 1, b\}$ with $b = k' + 1$ and $2 \leq b \leq m/2$ is affinely equivalent to $S_k$, then

$$b \in \{a, 1 - a\}.$$

But $1 - a \equiv m/2 + 2 \pmod{m}$, which is outside this range. Hence $b = a$ and $k' = k$. □

The addendum concludes that "all $P_{m,k}$ on 66 or fewer vertices are 2-edge-Hamiltonian" [2, p. 103]. Its table carries no entry at $(m, k) = (32, 14)$, although $2 < \gcd(33, 15) = 3 < 15$; by Proposition 3 the graph $P_{32,14}$ is alone in its isomorphism class, consistently with the count of 9 classes for $m = 32$ in [1, Table 3], and its 64 vertices put it beyond the 56 or fewer vertices reported in [1, p. 115]. Theorem 1 therefore supplies the one case needed for that 66-vertex assertion. The next pair of that residue in our family, $P_{56,26}$, lies beyond the addendum's range. The companion note [3] in this collection gives an independent covering certificate for $P_{32,14}$ and settles the isomorphism class $P_{34,9} \cong P_{34,14}$, which the addendum records as not known to be 2-edge-Hamiltonian.

## References

[1] G. J. Simmons, *Two related (?) 2-edge-Hamiltonian bigraph conjectures*, Bull. Inst. Combin. Appl. **88** (2020), 98–117.

[2] G. J. Simmons, *Two related (?) 2-edge-Hamiltonian bigraph conjectures addendum*, Bull. Inst. Combin. Appl. **89** (2020), 102–103.

[3] *Two residual cases of Simmons's polygonal bigraph conjecture*, companion note in this collection, file `residual-polygonal-bigraphs-close-simmons-finite-range` in `constructions/`.

[4] D. Wiedemann and M. E. Zieve, *Equivalence of sparse circulants: the bipartite Ádám problem*, arXiv:0706.1567, 2007.